

A. Setup (5–10 min)

Option 1: Quick & Simple (local SQLite + MongoDB Atlas free cluster)

1. **Install SQLite** (already comes with Python):

```
pip install sqlite-utils
```

2. **Install MongoDB Python driver:**

```
pip install pymongo
```

3. Create project folder:

```
de-section2-lab/
```

```
├── lab2_sqlite.py
```

```
├── lab2_mongo.py
```

```
└── schema_design.md
```

B. Relational Databases (RDBMS) – SQLite Mini Store (20–25 min)

lab2_sqlite.py

```
import sqlite3
```

```
# connect to database (file-based)
```

```
conn = sqlite3.connect("store.db")
```

```
cur = conn.cursor()
```

```
# create tables
```

```
cur.execute("""
```

```
CREATE TABLE IF NOT EXISTS customers(
```

```
customer_id INTEGER PRIMARY KEY,
```

```
name TEXT,
```

```
email TEXT UNIQUE
```

```
)
```

```
""")
```

```
cur.execute("""
```

```

CREATE TABLE IF NOT EXISTS products(
product_id INTEGER PRIMARY KEY,
name TEXT,
price REAL
)
"""
cur.execute("""
CREATE TABLE IF NOT EXISTS orders(
order_id INTEGER PRIMARY KEY,
customer_id INTEGER,
product_id INTEGER,
qty INTEGER,
order_date TEXT,
FOREIGN KEY(customer_id) REFERENCES customers(customer_id),
FOREIGN KEY(product_id) REFERENCES products(product_id)
)
""")

# insert sample data
cur.executemany("INSERT OR IGNORE INTO customers VALUES (?, ?, ?)", [
(1, "Alice", "alice@email.com"),
(2, "Bob", "bob@email.com"),
])
cur.executemany("INSERT OR IGNORE INTO products VALUES (?, ?, ?)", [
(1, "Laptop", 999.0),
(2, "Mouse", 25.5),
(3, "Keyboard", 49.9),
])
cur.executemany("INSERT OR IGNORE INTO orders VALUES (?, ?, ?, ?, ?)", [
(1, 1, 1, 1, "2024-01-01"),
(2, 2, 2, 2, "2024-01-02"),
])

conn.commit()

# queries
print("\nAll customers:")

```

```

for row in cur.execute("SELECT * FROM customers"):
    print(row)

print("\nJoin customers + orders + products:")
for row in cur.execute("""
SELECT c.name, p.name, o.qty, (o.qty * p.price) AS total
FROM orders o
JOIN customers c ON o.customer_id = c.customer_id
JOIN products p ON o.product_id = p.product_id
"""):
    print(row)

conn.close()

```

Run:

```
python lab2_sqlite.py
```

 You practiced **DDL (create)**, **DML (insert)**, and **queries with joins**.

C. NoSQL Databases — MongoDB Mini Store (25–30 min)

lab2_mongo.py

```

from pymongo import MongoClient

# connect to local or Atlas cluster
client = MongoClient("mongodb://localhost:27017/") # or Atlas URI
db = client["store_db"]

# collections
customers = db["customers"]
orders = db["orders"]

# reset collections
customers.drop()
orders.drop()

```

insert documents

```
customers.insert_many([
{"_id": 1, "name": "Alice", "email": "alice@email.com"},
{"_id": 2, "name": "Bob", "email": "bob@email.com"},
])
```

```
orders.insert_many([
{"_id": 1, "customer_id": 1, "items": [{"product": "Laptop", "qty": 1, "price": 999.0}]},
{"_id": 2, "customer_id": 2, "items": [{"product": "Mouse", "qty": 2, "price": 25.5}]},
])
```

queries

```
print("\nFind one customer:")
print(customers.find_one({"name": "Alice"}))
```

```
print("\nOrders for Bob:")
for o in orders.find({"customer_id": 2}):
print(o)
```

```
print("\nAggregate total revenue per customer:")
pipeline = [
{"unwind" : "$items"},
{"KaTeX parse error: Expected '}', got 'EOF' at end of input: ...oup": {"_id": "customer_id",
"total_spent": {"KaTeX parse error: Expected '}', got 'EOF' at end of input: sum": {"multiply":
["$items.qty", "$items.price"]}}}]
]
for res in orders.aggregate(pipeline):
print(res)
```

✓ You practiced **document inserts**, **find queries**, and **aggregation pipelines**.

D. Data Modeling & Schema Design (20–25 min)

Task

1. Design a **logical schema** for the e-commerce store in both SQL & NoSQL.
2. Save it in schema_design.md.

SQL Model → Normalized

- Customers (customer_id, name, email)
- Products (product_id, name, price)
- Orders (order_id, customer_id, order_date)
- OrderItems (order_id, product_id, qty, unit_price)

NoSQL Model → Embedded

```
{
  "_id": 101,
  "customer": { "id": 1, "name": "Alice" },
  "order_date": "2024-01-01",
  "items": [
    { "product": "Laptop", "qty": 1, "price": 999.0 },
    { "product": "Mouse", "qty": 1, "price": 25.5 }
  ]
}
```

✅ Compare: **SQL = normalized + joins**, **NoSQL = denormalized + embedded**.

E. What to Submit

1. Screenshot of **SQLite query output** (customers + joins).
 2. Screenshot of **MongoDB query output** (aggregation).
 3. schema_design.md with both relational + NoSQL schemas.
 4. Reflection: "Which model is easier for analytics? Which for scaling web apps?"
-

F. Stretch Goals (optional)

- Add indexing in SQLite and MongoDB.
 - Try MongoDB Atlas free cluster (cloud DB).
 - Visualize ERD (draw.io or dbdiagram.io).
 - Add a **JSON column** in SQLite to simulate hybrid storage.
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Mapping to Lectures

- **Lecture 4 (RDBMS)** → SQLite lab.
- **Lecture 5 (NoSQL)** → MongoDB lab.
- **Lecture 6 (Modeling & Schema Design)** → schema exercise & comparison.